

Tutorial 9 Rubric

Ans 1.

	1	2	3	4	5	6	7	8	9	10
ADD R1, R2, R3	F	D	E	W						
SUB R4, R1, R2		F	D	D	D	E	W			
B 5			F	F	F	D	E	W		
SUB R5, R6, R7						F	-	-	-	-
ADD R4, R5, R6							F	D	E	W

$$\text{CPI} = (10 \text{ Cycles}) / (5 \text{ Instructions}) = 2$$

Ans 2.

Dependence is a property of the instructions in a program; for example, a RAW dependence arises when one instruction uses the value produced by an earlier instruction.

A hazard is a potential problem in a pipeline that may arise from dependence. For example, a RAW dependency causes a hazard when the value produced by the earlier instruction is not yet available in the register file when the dependent instruction attempts to fetch it.

(Taken from Hannes and Patterson)

Ans 3.

As each instruction modifies 4 bytes and each memory location is 2 bytes, we would require two memory locations to store the value completely. Hence, the stack pointer value is given by:

1. PUSH 0xA02
2. POP 0xA00
3. POP 0x9FE
4. POP 0x9FC
5. PUSH 0x9FE
6. POP 0x9FC